

Definition and first terms

$13A(x) = 12 + 32x + A(x)^5$, with the origin-normalized solution.
 Normalization/grammar: $A(x)=1+(4)*T(x)$; $T=x+5*T^2+20*T^3+40*T^4+32*T^5$.
 $a(0), \dots, a(7) = 1, 4, 20, 280, 4660, 86728, 1727880, 36047280$.
 Kernel used throughout: $\rho(u) = u(1 - 5u^1 - 20u^2 - 40u^3 - 32u^4)$, so $x = \rho(T(x))$.

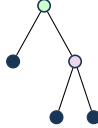
Typogeometry



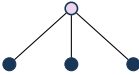
1x4



{1,1}x20



{1,{1,1}}x100



{1,1,1}x80

colored arity geometry; node color distinguishes constructors. Filled endpoints are true leaves; open endpoints are false/empty leaves. For colored grammars, color is genuine constructor data and is not silently converted into a positional zero pattern.

Typogeometric constructor codes and multiplicities: Δ_2 : 5, Δ_3 : 20, Δ_4 : 40, Δ_5 : 32.

Coefficient and contour extraction

$$\mathcal{K}_n = \{(k_1, \dots, k_4) \in \mathbb{Z}_{\geq 0}^4 : \sum_{j=1}^4 jk_j = n - 1\}, \quad K = \sum_{j=1}^4 k_j,$$

$$a(n) = \frac{4}{n} \sum_{k \in \mathcal{K}_n} \binom{n+K-1}{K} \binom{K}{k_1, \dots, k_4} 5^{k_1} 20^{k_2} 40^{k_3} 32^{k_4}.$$

$$a(n) = \frac{4}{2\pi i n} \oint_{\gamma} \frac{du}{\rho(u)^n}, \quad n \geq 1.$$

Recurrence

$\sum_{r=0}^4 P_r(n)a(n+r) = 0$ for $n \geq 1$. The exact degree-4 coefficient polynomials are embedded in `certificate_payload.json` (source: `data/recurrence.json`).

Differential equation

The scalar linear ODE has order 4. It is obtained from the recurrence by the standard Euler-operator substitution. Exact coefficients and boundary polynomial: `data/ode.json`.

Matrices, telescoper certificate, and checks

No compact matrix tuple is shown; see the embedded exact reduction data.

Certificate.

$R(n,u) = N(n,u)/\rho(u)^3$, $\deg_u N = 16$.
 Telescoping identity: $\sum_r P_r(n)H_{n+r}(u) = \partial_u(R(n,u)H_n(u))$.
 Matrix dimensions: 10×10 ; remainder 4×5 , rank 4, nullity 1. The exact telescoping residual is zero. The recurrence reconstructs all 24 stored terms; 6/6 published OEIS prefix terms match.